

Codimension-4 degenerations of nilpotent orbit closures and type AIII symmetric pairs

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AMS 2026 Fall Southeastern Sectional Meeting
Geometric and homological methods in representation theory
October 10–11, 2026

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- 1 Review of Kraft–Procesi minimal degenerations
- 2 Codimension-4 degenerations
- 3 Involutions of quiver varieties and symmetric pairs

Minimal degenerations

Nilpotent $\mathrm{GL}_n(\mathbb{C})$ -orbits are indexed by partitions $\lambda \vdash n$.

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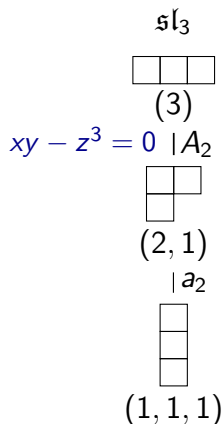
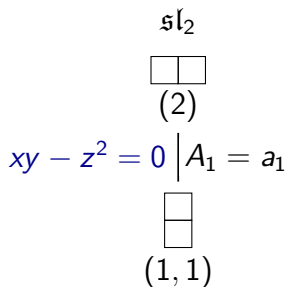
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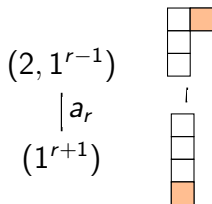
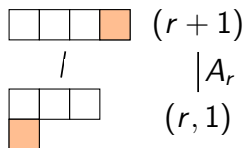
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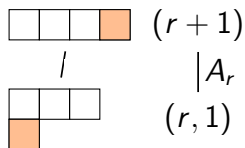
Hasse diagram in $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$



A_r/a_r singularity and row/column removal

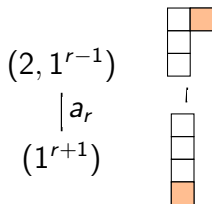


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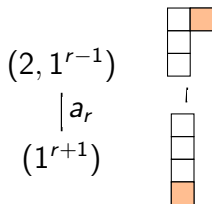
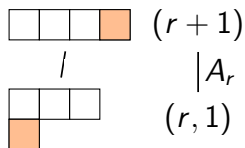
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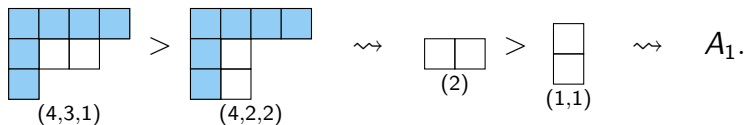
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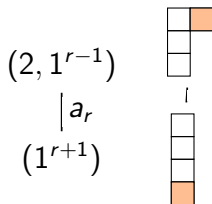
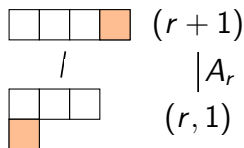
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Kraft-Procesi's removal of common row and columns (blue boxes)



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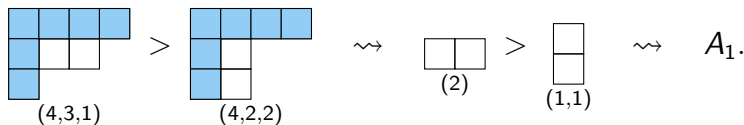
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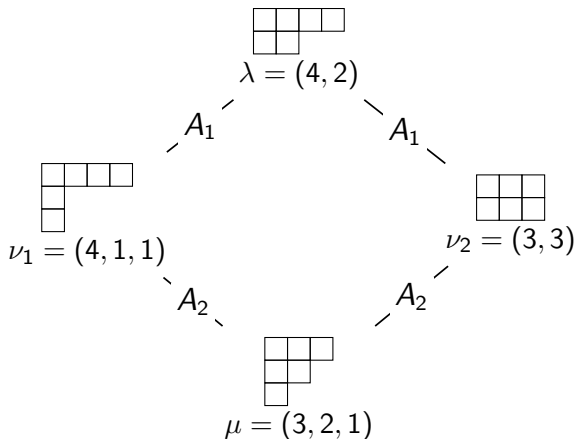


Codim-4 degenerations: $a_2 = \overline{\mathbb{O}}_{\min}(\mathfrak{sl}_3)$ or two steps of A_r 's.

Non-minimal degenerations

What happens when there are intermediate orbits ν occur?

$$\lambda = (4, 2), \quad \mu = (3, 2, 1), \quad \dim S_\mu \cap \overline{\mathbb{O}}_\lambda = 4.$$



What is the singularity of the entire 4d variety $S_\mu \cap \overline{\mathbb{O}}_\lambda$?

Combinatorial classification of codimension-4 degenerations

A combinatorial classification

For $\mu < \lambda$ with $\dim S_\mu \cap \overline{\mathbb{O}}_\lambda = 4$, after simultaneous row and column removals, the pair $\mu < \lambda$ is one of the following cases:

- 1 $(n) > (n-2, 2)$
- 2 $(m, n) > (m-1, n, 1)$
- 3 $(m, k, n) > (m-1, k+1, n-1, 1)$

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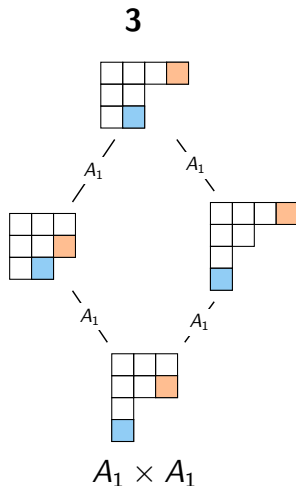
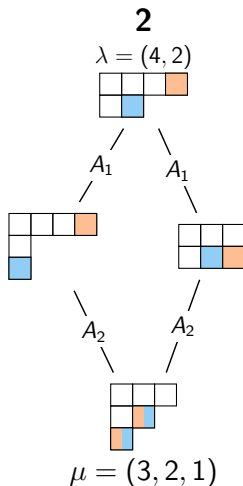
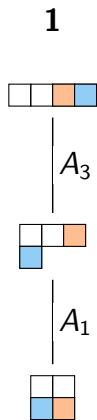
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Remark:

- In Case 1, $\overline{\mathbb{O}}_{(n)} = \mathcal{N}$, the nilpotent cone.
- In Case 3, $S_\mu \cap \overline{\mathbb{O}}_\lambda \simeq A_{m-k-1} \times A_{n-1}$.
- We focus on Case 2 today! Note that Case 2 includes the minimal nilpotent orbit closure $a_2 (2, 1) > (1, 1, 1)$ in $\mathfrak{sl}_3(\mathbb{C})$.

Three examples and their intermediate orbits



Ambient variety: $S_\mu \cap \overline{\mathcal{O}}_\lambda$ in Case 2

Proposition (H., in preparation)

In Case 2, we have

$$\mathbb{C}[S_\mu \cap \overline{\mathcal{O}}_\lambda] \simeq \frac{\mathbb{C}[x_1, x_2, x_3, y_1, y_2, y_3, z_1, z_2]}{I_2(M)},$$

where M is a 3×3 matrix whose entries are monomials of the generators, and $I_2(M)$ is the ideal generated by its 2×2 minors.

Examples of M :

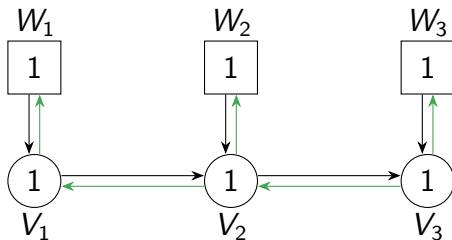
$$\begin{array}{cc} (4, 2) > (3, 2, 1) & \overline{\mathcal{O}}_{\min}(\mathfrak{sl}_3) = a_2 \\ \begin{pmatrix} z_1^2 & x_1 & x_3 \\ y_1 & -(z_1 + z_2) & x_2 \\ y_3 & y_2 & z_2^2 \end{pmatrix} & \begin{pmatrix} z_1 & x_1 & x_3 \\ y_1 & -(z_1 + z_2) & x_2 \\ y_3 & y_2 & z_2 \end{pmatrix} \end{array}$$

Proof technique: realize $S_\mu \cap \overline{\mathcal{O}}_\lambda$ as a quiver variety

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$$\Phi : \mathfrak{M}_0(\mathbf{v}, \mathbf{w}) \xrightarrow{\sim} S_\mu \cap \overline{\mathcal{O}}_\lambda.$$

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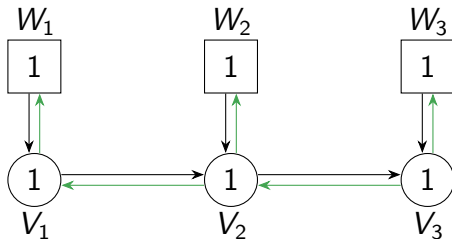


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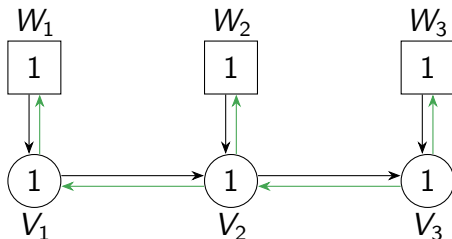
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$$\begin{aligned} z_1 : W_1 &\rightarrow W_1, & z_2 : W_3 &\rightarrow W_3 \\ x_1 : W_1 &\rightarrow W_2, & x_2 : W_2 &\rightarrow W_3, & x_3 : W_1 &\rightarrow W_3 \end{aligned}$$

Symmetric pairs and involutions of quiver varieties

For an involution $\sigma: \mathfrak{g} \rightarrow \mathfrak{g}$, set $\theta := -\sigma$. Write

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \quad \mathfrak{k} = \mathfrak{g}^\sigma, \quad \mathfrak{p} = \mathfrak{g}^\theta.$$

Type	Symmetric pair $(\mathfrak{g}, \mathfrak{k})$
AI	$(\mathfrak{sl}_n, \mathfrak{so}_n)$
AI	$(\mathfrak{sl}_n, \mathfrak{sp}_n)$, n even
AIII	$(\mathfrak{sl}_n, \mathfrak{s}(\mathfrak{gl}_p \oplus \mathfrak{gl}_q))$, $p + q = n$

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$$\Phi \circ \theta_Q = \theta \circ \Phi,$$

$$\mathfrak{M}_0(\mathbf{v}, \mathbf{w})^{\theta_Q} \simeq (S_\mu \cap \overline{\mathcal{O}}_\lambda)^\theta = S_\mu \cap \overline{\mathcal{O}}_\lambda \cap \mathfrak{p}.$$

when θ arises from a symmetric pair of types **AI** and **AI**.

AIII involution and the Case 2 fixed-point locus theorem

In work in preparation, I construct an involution of quiver varieties by *assigning signatures to the framings*, together with appropriate sign changes of the quiver maps, realizing the **AIII** symmetric pair.

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In Case 2, the fixed-point locus $(S_\mu \cap \overline{\mathcal{O}}_\lambda)^\theta$ is a union of copies of $\mathbb{C}^2$, meeting along lines or at points.

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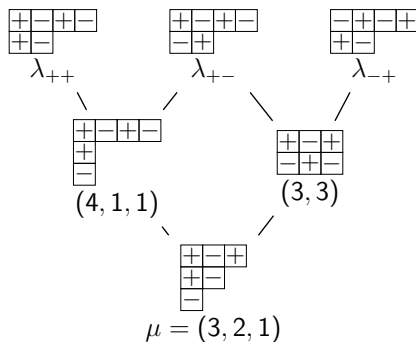
Remark: The AIII involution of quiver varieties construction work for any dimension and framing vectors $\mathbf{v}, \mathbf{w}$. For the fixed-point loci we care about, we just applied it to the 4d case.

The signed Hasse diagram: $(4, 2) > (3, 2, 1)$

For the symmetric pair $\mathfrak{sl}_n(\mathbb{C}), \mathfrak{s}(\mathfrak{gl}_p(\mathbb{C}) \oplus \mathfrak{gl}_q(\mathbb{C}))$, nilpotent K -orbits in $\mathfrak{p}$ are parametrized by signed Young diagrams: $\#p +$, $\#q -$ in total; signs alternate in each row.

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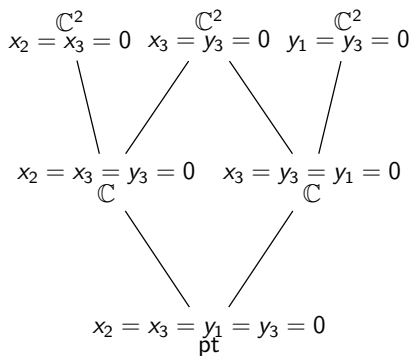
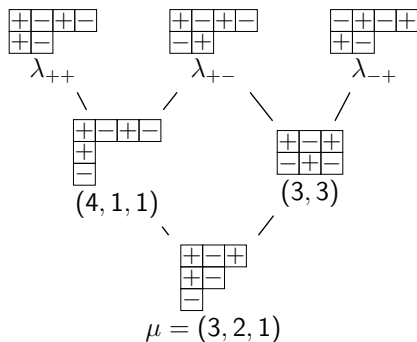
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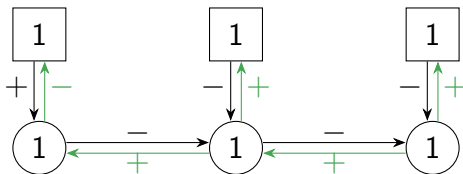
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$$(S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta \simeq \text{Spec } \mathbb{C}[x_2, x_3, y_2, y_3] / (x_2 y_3, x_3 y_3, x_3 y_2)$$



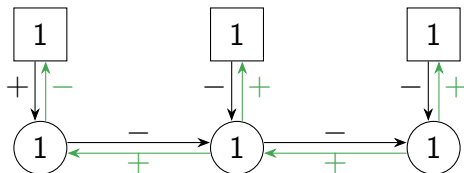
Example of involution of quiver varieties

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We have $\theta(z_i) = -z_i$; $\theta(x_1) = -x_1$, $\theta(y_2) = -y_2$, resulting in that

$$(S_\mu \cap \overline{\mathbb{O}})^\theta \simeq \text{Spec } \mathbb{C}[x_2, x_3, y_1, y_3] / (x_2 y_3, x_3 y_3, x_3 y_1).$$

Thank you!